

B1690-A MRC



2689 Martin Luther King Jr Blvd
Youngstown, OH 44510

VALLOUREC STAR LP
Seamless Tubular Products

Certified Test Report
12/30/2024
Revision: 0

Order: Y-221128255-001 Desc: Seamless Line Pipe - As Rolled

Customer: MRC Global (US) Inc.

SLN / (Heat): CND211

Grade: API SL X42/B N/BN PSL2 46th Edition May 01, 2019

Customer Order: S 032 656823

Split: 01

OD(in): 4.500

Wall(in): 0.337

Lbs/Ft: 15.00

Length: Double Random

End Finish: Plain End Beveled

Comments: Melted and Manufactured in Youngstown, Ohio USA

Pipe is also manufactured to: ASTM A106B/C-19a, ASTM A53B-24, ASME SA106B/C-2023, ASME SA53B-2023. Vallourec Star's QA program meets the requirements of Directive 2014/68/EU and DIN EN10204-2004 Type 3.1. Satisfies NACE MR0175/ISO15156:2015 and ANSI/NACE MR0103/ISO17945:2015. Material has been produced using silicon killed carbon steel with aluminum fine grain practice. No repair welding has been conducted. Material is free from Mercury.

Mechanical Properties

		Tensile Size			Strength (ksi)			Elongation		
Orientation	Type	Specimen Cross Section			Yield	Tensile	Y/T Ratio	Gage Length	% Elong	% RA
Longitudinal	Strip	Width (in)	Thick (in)	Area (sqin)	50.9	78.5	0.65	2	35.4	46.8

Required Min/Max Yield Strength (ksi): 42.1/71.8

Required Min/Max Tensile Strength (ksi): 70/95

Calculated Max Internal Yield (psi): 6667

Required Min Elongation(%): 30

Supplemental Requirements

Flattening Test	Passed
Hardness 3 Point	78.5 avg HREBW
Hydrostatic Pressure	4730 psi
Hydrostatic Hold Time	5 seconds

Chemical Analysis

Chem Spec: 66S

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
Heat	0.19	1.07	0.008	0.004	0.24	0.16	0.06	0.11	0.02	0.009	0.000	0.003	0.024	0.0018	0.0003	0.002	0.0072	0.41
Product 1	0.19	1.06	0.011	0.004	0.24	0.15	0.06	0.11	0.03	0.006	0.000	0.002	0.025	0.0021	0.0002	0.002	0.0065	0.41
Product 2	0.20	1.06	0.011	0.004	0.24	0.15	0.06	0.11	0.03	0.007	0.000	0.002	0.024	0.0019	0.0001	0.002	0.0075	0.41

Charpy Impact Testing - Not required per API.

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star without the expressed written consent of Vallourec Star negates the validity of this test report. Vallourec Star is not responsible for the inability of this material to meet specific applications.

THIS CERTIFICATE IS NOTARIZED ONLY WHEN REQUESTED.

Signed: *Jeff Schultz*

Name: Jeff Schultz, Lab Supervisor

Date Approved: 12/30/2024

This document is uncontrolled when printed. Always refer to the MTR System for the most current version. FS-VMS-8.2.4.1.1 Rev 7



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Inspection Company: L Bay

Non-Destructive Evaluation Criteria

UT CHANNELS

ID

10% Longitudinal

10% Transverse

OD

5% Longitudinal

5% Transverse

WALL THICKNESS CHECK

87.5% Wall Reduction

Required Volumetric Coverage: 100%

ADDITIONAL REQUIREMENTS

Notch Length 1.5" Max

Notch Width .020" Max

Untested Ends Cropped

OD

Visual Insp. Full Length - White Light

ID

Visual Insp. Pipe Ends - White Light